

Field Oriented Control (FOC) for a 3-Phase Induction Motor

*Project Report Submitted in partial fulfilment of the requirements for the degree
of Bachelor of Technology from Maulana Abul Kalam Azad University of
Technology, West Bengal (formerly known as West Bengal University of
Technology)*

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Certificate of Recommendation

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Board of Examiners

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Acknowledgement

We would like to express our profound gratitude to our project guide, Prof. Deepjyoti Santra, for his valuable guidance, continuous support, and encouragement throughout the course of this project. His deep knowledge and technical insights into electrical drives and vector control have been instrumental in shaping this analysis.

We are also deeply thankful to Prof. Atanu Maji, Head of the Electrical Engineering Department, for providing us with the necessary academic framework, laboratory facilities, and a conducive environment to carry out this work successfully.

We also extend our heartfelt thanks to the faculty, staff, and administration of Cooch Behar Government Engineering College for their continuous assistance and cooperation.

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Abstract

This report provides a comprehensive, component-level technical analysis of the Simulink model `VECTOR_CONTROL_FOC_OF_IM_2023.slx`, which implements an Indirect Field Oriented Control (IFOC) architecture for a 50 HP Squirrel-Cage Induction Motor. Intended to bridge complex electromechanical theory with practical digital control implementation, this report systematically deconstructs the frameworks embedded within the simulation. It establishes the plant's equivalent circuit parameters and models the dynamic transient response of the decoupling matrices.

The analysis explores the critical coordinate transformations (Forward and Reverse Clarke and Park) required to map stationary three-phase variables into a synchronously rotating $d - q$ reference frame. By meticulously evaluating the system's decoupling equations, the report demonstrates how IFOC achieves independent, high-performance regulation of rotor flux and electromagnetic torque, mirroring the dynamics of a separately excited DC machine. Furthermore, the discrete-time closed-loop control strategy is examined, detailing the tuning parameters of the cascaded Proportional-Integral (PI) controllers and slip estimation. Finally, the report dissects the Space Vector Pulse Width Modulation (SVPWM) synthesis used to drive the three-phase inverter, verifying its advantages in optimising DC bus voltage utilisation and minimising total harmonic distortion.

(i_a, i_b, i_c) are projected onto a two-dimensional, synchronously rotating reference frame ($d - q$ axes).

By deliberately aligning the direct-axis (d -axis) of this rotating frame with the rotor flux linkage vector, the overall stator current is mathematically decomposed into two entirely independent components:

- **Flux-Producing Component (i_{ds}):** The direct-axis stator current is constrained to align perfectly with the rotor flux linkage vector (ψ_r). It is strictly responsible for establishing and maintaining the internal magnetisation of the machine.
- **Torque-Producing Component (i_{qs}):** The quadrature-axis stator current is maintained precisely at a 90° electrical angle relative to the rotor flux axis. Because the entirety of the machine's flux is mathematically isolated onto the d-axis ($\psi_{qr} = 0$), the torque generation relies exclusively on this component.

1.2 Torque Calculation and Decoupling

The developed electromagnetic torque can be expressed via the decoupled equation:

$$T_e = \frac{3}{2} \left(\frac{P}{2} \right) \frac{L_m}{L_r} \psi_r i_{qs} \quad (1.1)$$

Based on the parameters defined for the 50 HP motor:

- Number of poles, $P = 4$
- Magnetizing inductance, $L_m = 34.7 \text{ mH} = 0.0347 \text{ H}$
- Rotor inductance, $L_r = L_{lr} + L_m = 0.8 \text{ mH} + 34.7 \text{ mH} = 35.5 \text{ mH} = 0.0355 \text{ H}$
- Nominal flux, $\psi_r = 0.96 \text{ Wb}$

Executing the calculation yields the direct torque constant:

$$\begin{aligned} T_e &= \frac{3}{2} \left(\frac{4}{2} \right) \left(\frac{0.0347}{0.0355} \right) (0.96) \cdot i_{qs} \\ T_e &= 3 \times (0.9775) \times (0.96) \cdot i_{qs} \\ T_e &\approx 2.8152 \cdot i_{qs} \text{ N} \cdot \text{m} \end{aligned} \quad (1.2)$$

This confirms a direct, linear torque constant of 2.8152 Nm/A for the q-axis current. Because the rotor flux ψ_r is held constant by the independent d-axis control loop, the electromagnetic torque T_e becomes directly, linearly, and instantaneously proportional to i_{qs} , eliminating cross-coupling dynamics.

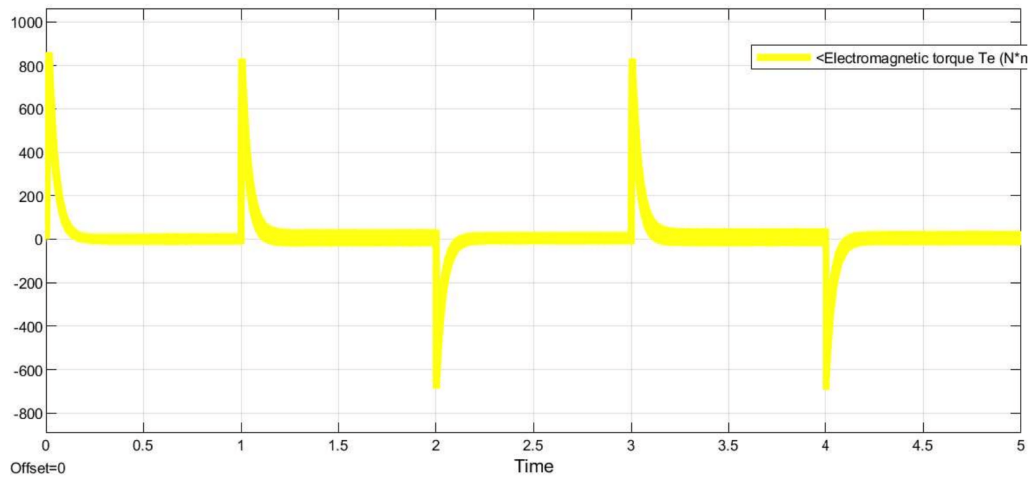


Figure 1.2: Scope Output: Transient and Steady-State Electromechanical Torque (T_e) Tracking the Load Demands

Chapter 2

Induction Motor Plant Parameters and Dynamic Modelling

2.1 Plant Parameters

The electromechanical plant governed by this control architecture is a three-phase Squirrel-Cage Induction Motor (SCIM). Because the rotor currents and rotor flux linkages are physically inaccessible and cannot be directly measured, the control strategy relies entirely on an accurate mathematical model of the machine.

Table 2.1: Squirrel-Cage Induction Motor Nominal Ratings and Equivalent Circuit Parameters

Symbol	Parameter Description	Value
Nominal Ratings		
S	Nominal Power	50 HP (37.3 kW)
V_s	Stator Line-to-Line Voltage	460 V
f	Nominal Frequency	50 Hz
P	Number of Poles	4 (2 Pole Pairs)
N_s	Synchronous Speed	1500 RPM
Electrical Parameters (Per Phase)		
R_s	Stator Resistance	0.087 Ω
R_r	Rotor Resistance (Referred to Stator)	0.228 Ω
L_{ls}	Stator Leakage Inductance	0.1 mH
L_{lr}	Rotor Leakage Inductance (Referred)	0.8 mH
L_m	Magnetizing Inductance	34.7 mH
Mechanical Parameters		
J	Moment of Inertia	0.662 kg·m ²
F	Viscous Friction Coefficient	0.1 N·ms/rad

2.2 Significance of Electrical Parameters in FOC

From the raw parameters provided in Table 2.1, the lumped self-inductances of the stator (L_s) and rotor (L_r) are derived as follows:

$$L_s = L_{ls} + L_m = 0.1 \text{ mH} + 34.7 \text{ mH} = 34.8 \text{ mH} = 0.0348 \text{ H} \quad (2.1)$$

$$L_r = L_{lr} + L_m = 0.8 \text{ mH} + 34.7 \text{ mH} = 35.5 \text{ mH} = 0.0355 \text{ H} \quad (2.2)$$

A critical derived parameter for Indirect FOC is the rotor time constant (τ_r). This value dictates the feedforward slip frequency calculation:

$$\tau_r = \frac{L_r}{R_r} = \frac{0.0355}{0.228} = 0.15570175 \text{ s} \approx 0.1557 \text{ s} \quad (2.3)$$

Furthermore, the machine's total leakage factor (σ), which dictates the maximum rate of change of stator currents (and thus the necessary bandwidth of the inner PI current controllers), is calculated as:

$$\begin{aligned} \sigma &= 1 - \frac{L_m^2}{L_s L_r} \\ \sigma &= 1 - \frac{(0.0347)^2}{(0.0348 \times 0.0355)} \\ \sigma &= 1 - \frac{0.00120409}{0.0012354} \\ \sigma &= 1 - 0.97465 = 0.02535 \end{aligned} \quad (2.4)$$

This highly coupled magnetic relationship ($\sigma \ll 1$) confirms a well-designed motor but necessitates high-gain, fast-acting current loops to force the stator currents to track the instantaneous FOC references.

2.3 Mechanical Dynamics and Speed Loop Tuning

The mechanical equation of motion that dictates the rotor's dynamic trajectory is governed by:

$$T_e - T_L = J \frac{d\omega_m}{dt} + F\omega_m \quad (2.5)$$

Substituting the mechanical parameters of the 50 HP motor ($J = 0.662$ and $F = 0.1$):

$$T_e - T_L = 0.662 \frac{d\omega_m}{dt} + 0.1 \cdot \omega_m \quad (2.6)$$

Assuming the load torque $T_L = 0$ for the primary plant transfer function analysis, taking

the Laplace transform yields:

$$\frac{\omega_m(s)}{T_e(s)} = \frac{1}{0.662s + 0.1} \quad (2.7)$$

This first-order transfer function reveals that the mechanical plant acts as a natural low-pass filter. The relatively large inertia (0.662) naturally attenuates high-frequency torque ripples generated by the SVPWM inverter, allowing the outer speed control loop to be tuned with robust Proportional-Integral (PI) parameters without reacting to switching noise.

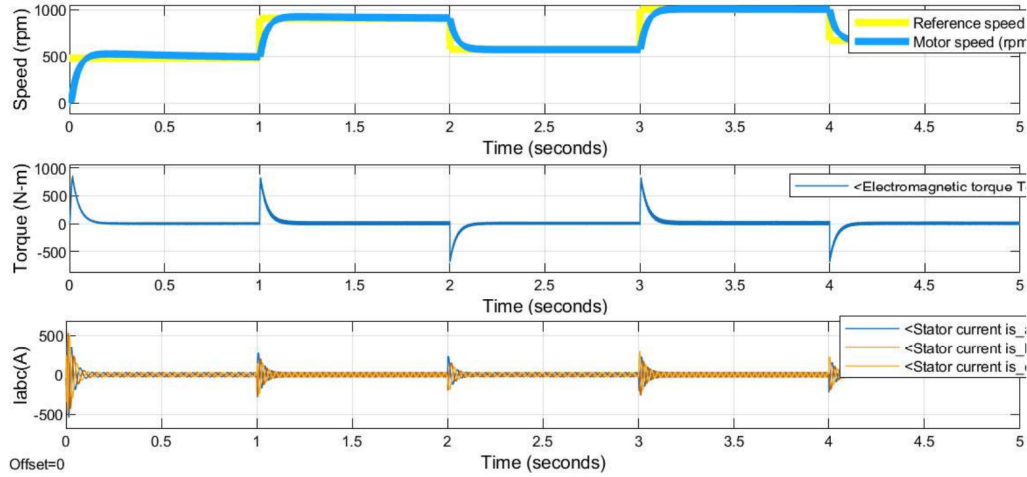


Figure 2.1: Scope Output: Overall Mechanical Dynamics (Speed Tracking, Torque, and Stator Currents)

Chapter 3

Control System Architecture and Discrete-Time Execution

The Simulink model is architected around a series of highly deterministic, cascading subsystems executing the Indirect Field Oriented Control (IFOC) algorithm. A critical parameter governing this digital control system is the fundamental sample time, set to $T_s = 2\mu s$. This exceptionally fast execution rate ensures stability in the high-bandwidth inner current loops.

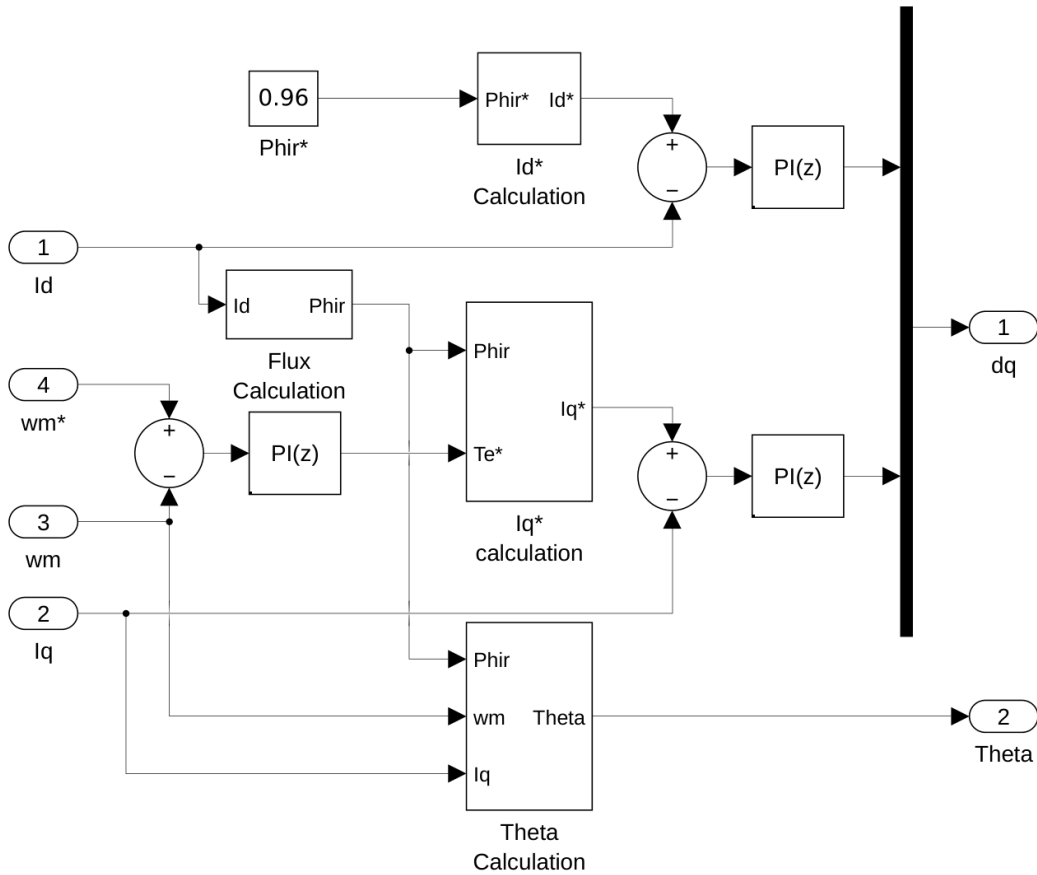


Figure 3.1: Detailed Implementation of the FOC Control Loops (Current and Flux Regulation)

3.1 Coordinate Transformations

The central premise of FOC lies in transforming the time-varying, three-phase AC stator variables into a synchronously rotating $d - q$ reference frame.

3.1.1 Forward Clarke Transform (Stationary $3 \rightarrow 2$ Projection)

The three-phase AC system is projected onto an equivalent two-dimensional orthogonal reference frame ($\alpha - \beta$ axes) using the amplitude-invariant Clarke matrix:

$$\begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} = \frac{2}{3} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} \quad (3.1)$$

Forward Clarke Transformation

$$\alpha(t) = \frac{2}{3}a(t) - \frac{1}{3}b(t) - \frac{1}{3}c(t)$$

$$\beta(t) = \frac{1}{\sqrt{3}}b(t) - \frac{1}{\sqrt{3}}c(t)$$

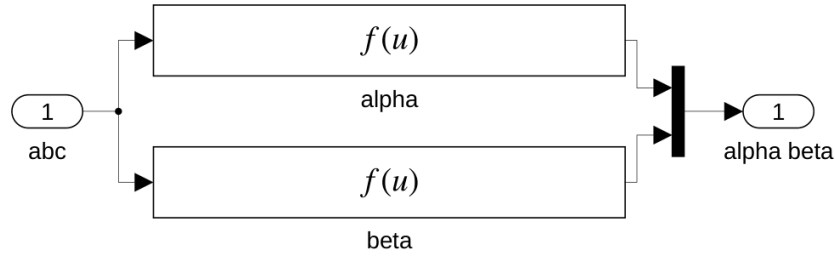


Figure 3.2: Simulink Subsystem: Forward Clarke Transform (abc to $\alpha\beta$)

3.1.2 Forward Park Transform (Stationary $\alpha - \beta$ to Rotating $d - q$)

To allow PI controllers to track signals without steady-state error, the stationary $\alpha - \beta$ frame is projected onto a synchronously rotating reference frame ($d - q$ axes) spinning at the electrical angular velocity of the rotor magnetic flux:

$$\begin{bmatrix} i_d \\ i_q \end{bmatrix} = \begin{bmatrix} \cos \theta_e & \sin \theta_e \\ -\sin \theta_e & \cos \theta_e \end{bmatrix} \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} \quad (3.2)$$

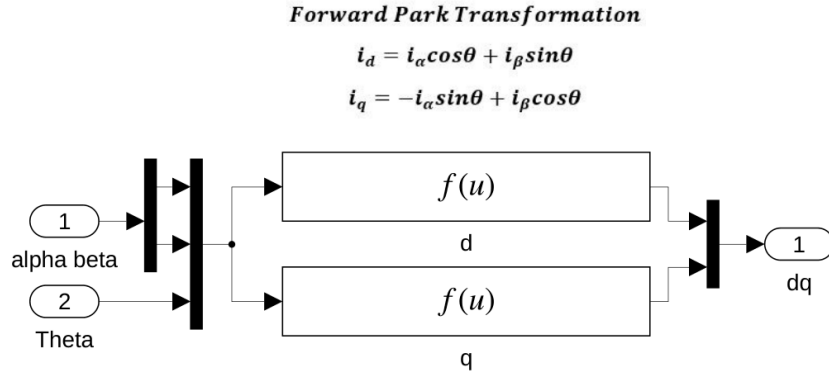


Figure 3.3: Simulink Subsystem: Forward Park Transform ($\alpha\beta$ to dq)

3.1.3 Reverse Park Transform

The PI current regulators output decoupling voltage commands (v_d^* and v_q^*). The Reverse Park transform performs the coordinate de-rotation back to the stationary frame for the SVPWM block:

$$\begin{bmatrix} v_\alpha^* \\ v_\beta^* \end{bmatrix} = \begin{bmatrix} \cos \theta_e & -\sin \theta_e \\ \sin \theta_e & \cos \theta_e \end{bmatrix} \begin{bmatrix} v_d^* \\ v_q^* \end{bmatrix} \quad (3.3)$$

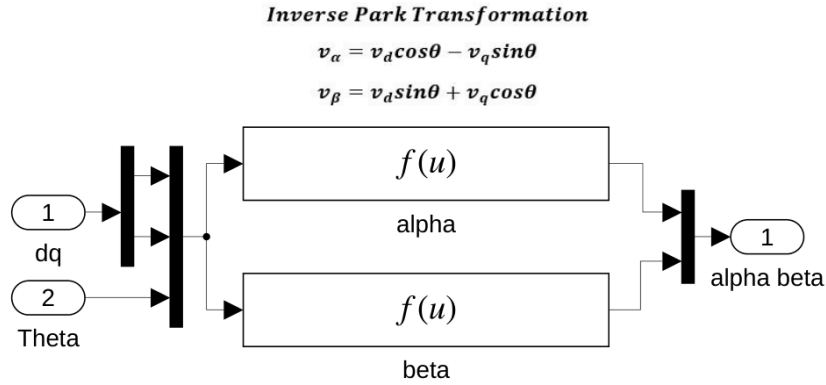


Figure 3.4: Simulink Subsystem: Reverse Park Transform (dq to $\alpha\beta$)

3.2 Flux and Torque Decoupling Equations

3.2.1 Direct-Axis Current Reference (i_{ds}^*)

The transient behaviour of the rotor flux linkage ψ_r is governed by:

$$\tau_r \frac{d\psi_r}{dt} + \psi_r = L_m i_{ds} \quad (3.4)$$

Under the steady-state assumption ($\frac{d\psi_r}{dt} = 0$), the reference direct-axis current simplifies to:

$$i_{ds}^* = \frac{\psi_r^*}{L_m} = \frac{0.96}{0.0347} = 27.6657 \text{ A} \quad (3.5)$$

The d-axis current controller constantly regulates to 27.67 A to maintain nominal flux.

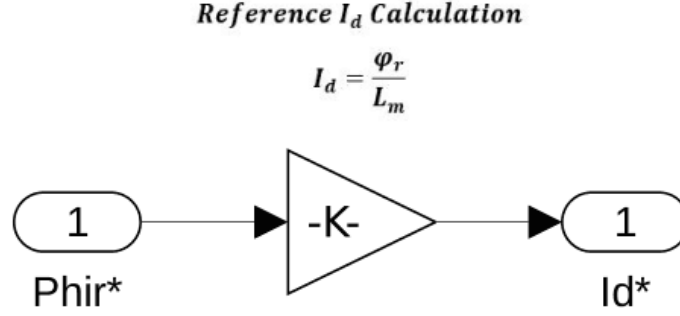


Figure 3.5: Simulink Subsystem: I_d^* Reference Calculation

3.2.2 Quadrature-Axis Current Reference (i_{qs}^*)

Aligning the d-axis perfectly with the rotor flux vector forces $\psi_{qr} = 0$. The torque equation becomes:

$$T_e = \frac{3}{2} \left(\frac{P}{2} \right) \frac{L_m}{L_r} \psi_r i_{qs} \quad (3.6)$$

Inverting this to solve for the necessary reference current i_{qs}^* :

$$\begin{aligned} i_{qs}^* &= \frac{2}{3} \cdot \frac{2}{P} \cdot \frac{L_{lr} + L_m}{L_m} \cdot \frac{T_e^*}{\psi_r} \\ i_{qs}^* &= \left(\frac{2}{3} \right) \left(\frac{2}{4} \right) \left(\frac{0.0355}{0.0347} \right) \frac{T_e^*}{0.96} \\ i_{qs}^* &= (0.3333) \times (1.02305) \times \frac{T_e^*}{0.96} = 0.3552 \cdot T_e^* \text{ A} \end{aligned} \quad (3.7)$$

To prevent a division-by-zero singularity during startup when $\psi_r \approx 0$, the digital implementation adds a numerical perturbation ($\epsilon = 1 \times 10^{-3}$):

$$i_{qs}^* \propto \frac{T_e^*}{\psi_r + 10^{-3}} \quad (3.8)$$

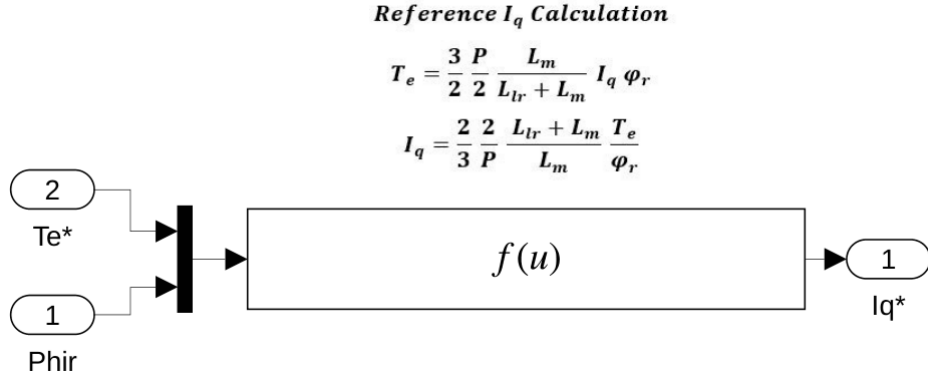


Figure 3.6: Simulink Subsystem: I_q^* Reference Calculation

3.3 Slip and Flux Angle (θ_e) Synthesis

Indirect FOC synthesises the flux angle mathematically in a feedforward manner by calculating the required slip frequency (ω_{sl}):

$$\omega_{sl} = \frac{L_m}{\tau_r} \cdot \frac{i_{qs}^*}{\psi_r^*} = \frac{0.0347}{0.1557} \cdot \frac{i_{qs}^*}{0.96} = 0.232 \cdot i_{qs}^* \text{ rad/s} \quad (3.9)$$

The synchronous speed (ω_e) is found by adding the slip to the electrical rotor speed ($\omega_r = \omega_m \frac{P}{2}$):

$$\omega_e = \omega_r + \omega_{sl} = 2\omega_m + 0.232 \cdot i_{qs}^* \text{ rad/s} \quad (3.10)$$

The instantaneous rotor flux angle is obtained via discrete integration ($T_s = 2\mu\text{s}$):

$$\theta_e[k] = (\theta_e[k-1] + \omega_e[k] \cdot T_s) \pmod{2\pi} \quad (3.11)$$

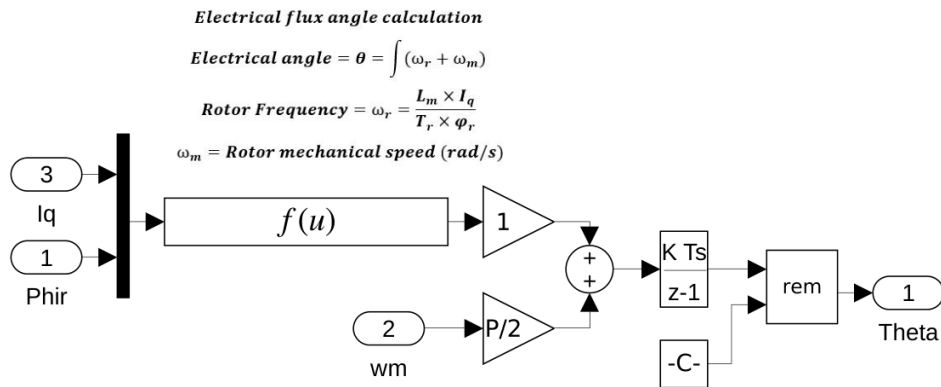


Figure 3.7: Simulink Subsystem: IFOC Slip and Flux Angle Estimation

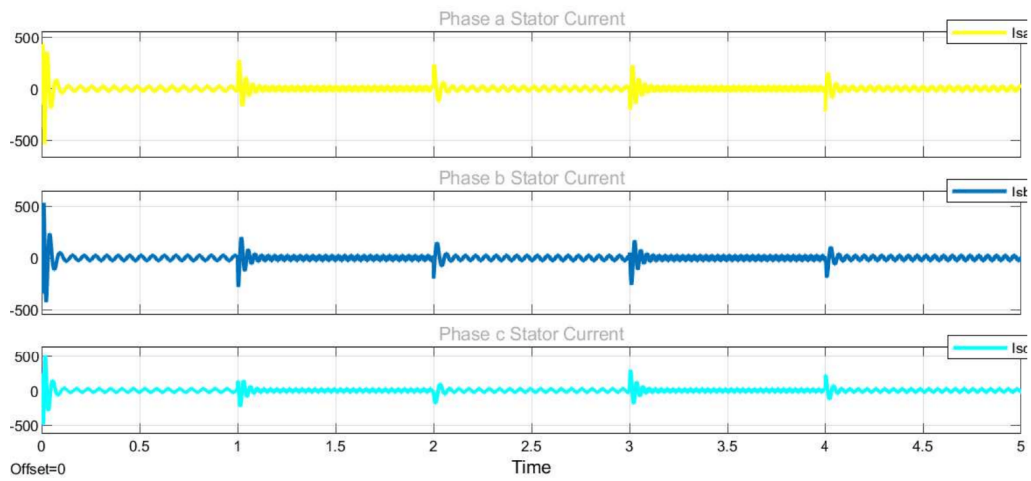


Figure 3.8: Scope Output: Three-Phase Stator Currents (I_a , I_b , I_c) Responding to Load Transients

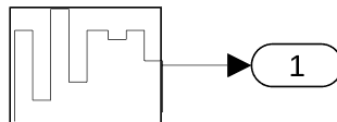


Figure 3.9: Reference Signal Builder Logic

Chapter 4

Space Vector Pulse Width Modulation (SVPWM)

The final control actuation is executed by a Three-Phase Voltage Source Inverter, powered by a 780 V DC link. The Space Vector PWM (SVPWM) subsystem translates continuous domain v_α^* and v_β^* voltage commands into discrete transistor switching signals, optimising DC bus utilisation by up to 15.47% over standard SPWM and significantly reducing total harmonic distortion.

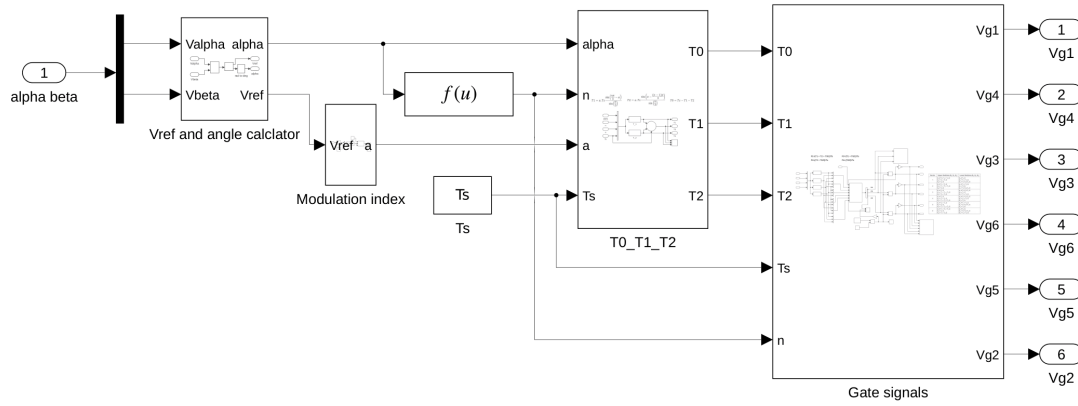


Figure 4.1: Main Architecture of the Space Vector PWM Generation Subsystem

4.1 SVPWM Algorithm Execution

4.1.1 Reference Vector Synthesis and Modulation Index

The model computes the magnitude and angle of the reference vector:

$$|\vec{V}_{ref}| = \sqrt{(v_\alpha^*)^2 + (v_\beta^*)^2} \quad (4.1)$$

$$\theta_{ref} = \arctan\left(\frac{v_\beta^*}{v_\alpha^*}\right) \quad (4.2)$$

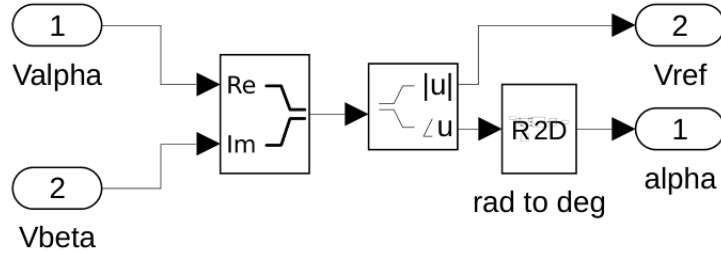


Figure 4.2: Simulink Subsystem: Transformation to Magnitude (V_{ref}) and Angle

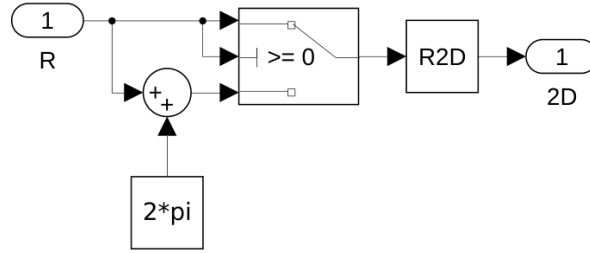


Figure 4.3: Simulink Subsystem: Angle Normalization (rad to deg)

To prevent non-linear operation, the modulation index a is bounded:

$$a = \frac{\sqrt{3} \cdot |\vec{V}_{ref}|}{V_{dc}} \leq 0.866 \quad (4.3)$$

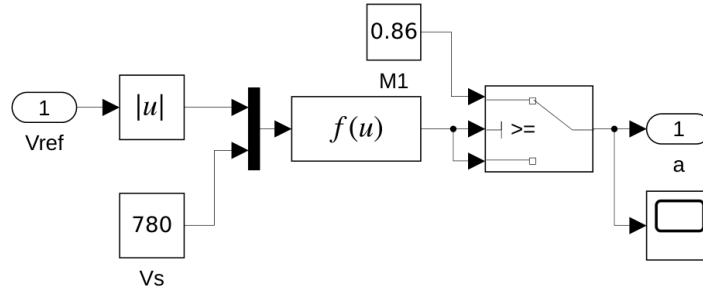


Figure 4.4: Simulink Subsystem: Modulation Index Calculation

4.1.2 Dwell Time Calculation

The dwell times for the active vectors are calculated using the volt-second balance principle over switching period T_s :

$$T_1 = a \cdot T_s \cdot \frac{\sin\left(\frac{n\pi}{3} - \alpha\right)}{\sin\left(\frac{\pi}{3}\right)} \quad (4.4)$$

$$T_2 = a \cdot T_s \cdot \frac{\sin\left(\alpha - \frac{(n-1)\pi}{3}\right)}{\sin\left(\frac{\pi}{3}\right)} \quad (4.5)$$

$$T_0 = T_s - T_1 - T_2 \quad (4.6)$$

$$T1 = a \cdot Ts \frac{\sin\left(\frac{n\pi}{3} - \alpha\right)}{\sin\left(\frac{\pi}{3}\right)} \quad T2 = a \cdot Ts \frac{\sin\left(\alpha - \frac{(n-1)\pi}{3}\right)}{\sin\left(\frac{\pi}{3}\right)} \quad T0 = Ts - T1 - T2$$

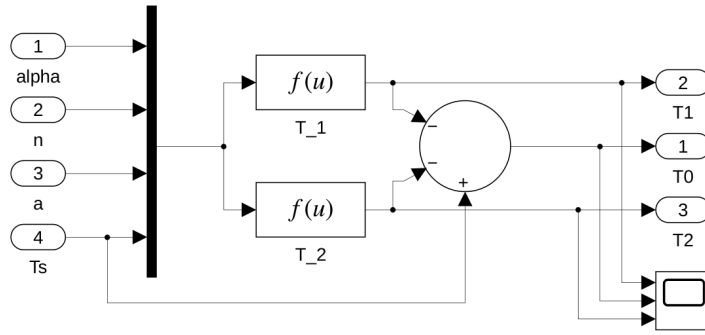


Figure 4.5: Simulink Subsystem: Dwell Time (T_1, T_2, T_0) Calculation

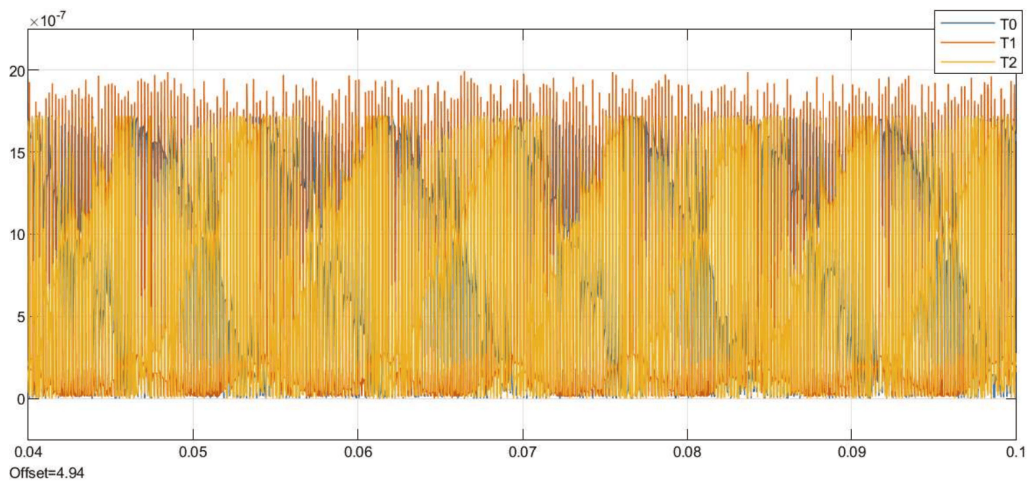


Figure 4.6: Scope Output: Calculated High-Frequency Dwell Times (T_0, T_1, T_2)

4.1.3 Switching Sequence Generation

The calculated dwell times are mapped to logical duty cycle values (F_1, F_2, F_3, F_4) via a multipoint switch governed by the sector number n , which are then compared against a high-frequency triangular carrier to generate physical Boolean pulses (V_{g1} to V_{g6}) for the IGBT drivers. A symmetrical switching pattern is employed.

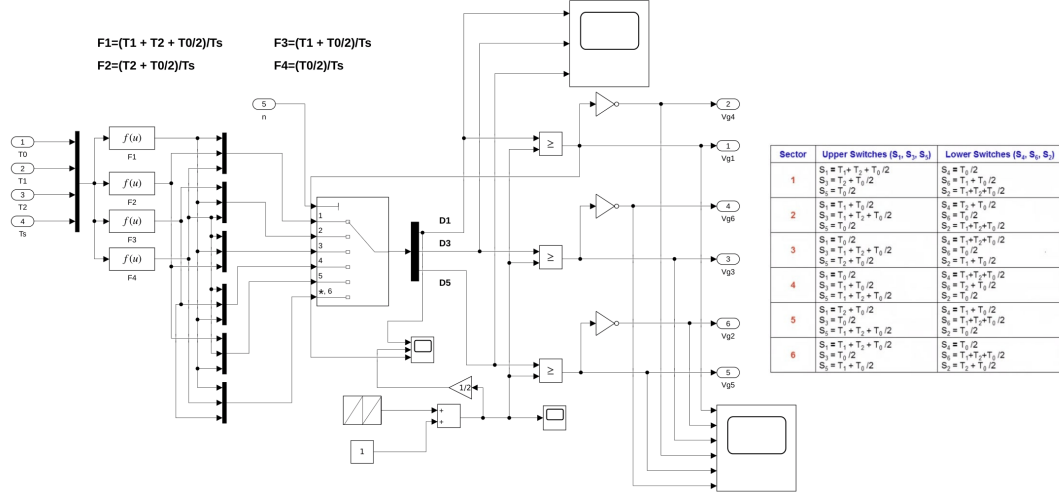


Figure 4.7: Simulink Subsystem: Detailed PWM Logic and Gate Signal Comparator Routing

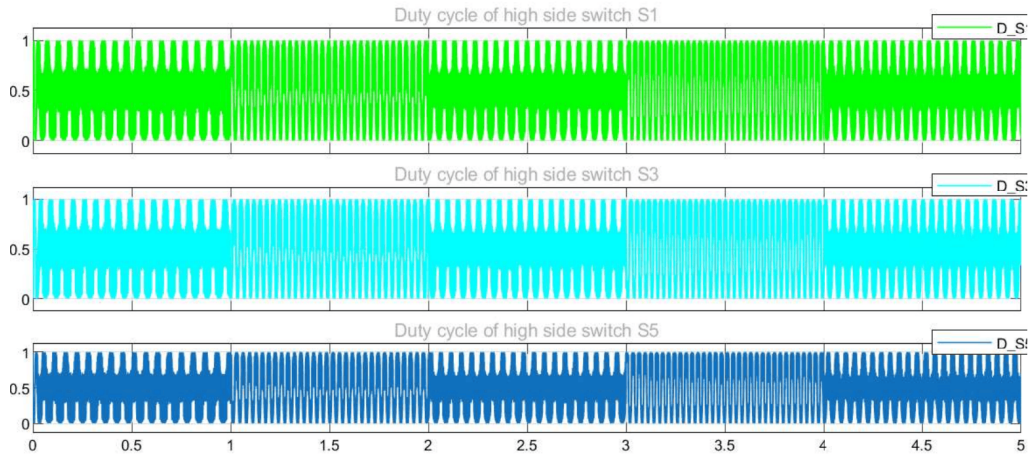


Figure 4.8: Scope Output: Duty Cycles of the High-Side Inverter Switches S1, S3, S5

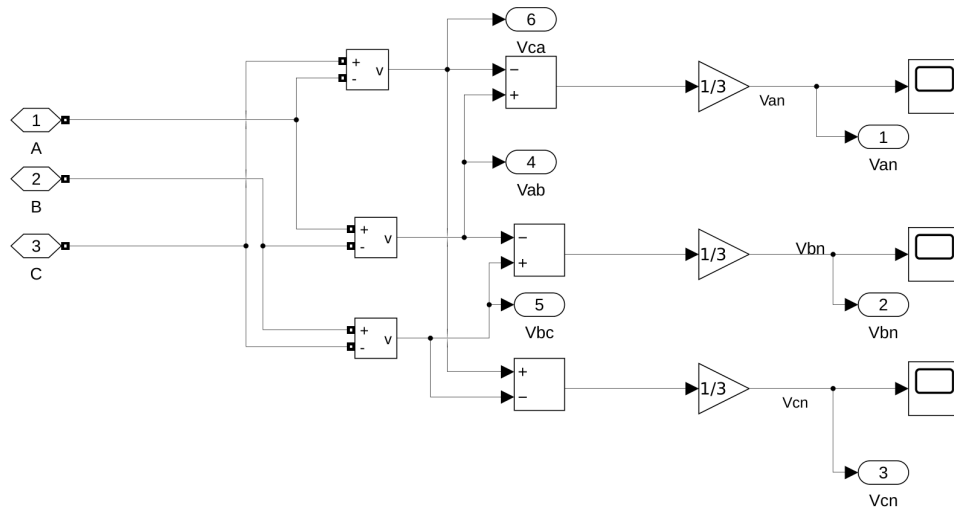


Figure 4.9: Simulink Subsystem: Three-Phase and Line-to-Line Inverter Voltage Calculation

Chapter 5

Conclusion and Future Scope

5.1 Conclusion

The presented Simulink model serves as a highly accurate, functionally complete digital twin of a modern, industrial-grade Indirect Field Oriented Control (IFOC) system for a Squirrel-Cage Induction Motor. By meticulously modelling the electromechanical plant, the coordinate transformations, and the discrete-time control logic, this architecture successfully demystifies the complex, non-linear dynamics inherent in three-phase AC machines.

The mathematical projection of the stator currents into a synchronously rotating $d - q$ reference frame proves that an induction motor can be controlled with the same orthogonal precision, and dynamic transient response, as a separately excited DC motor. Operating at a stringent $2\mu s$ sample time, the model faithfully replicates the execution environment of a modern Digital Signal Processor (DSP), highlighting the necessary bandwidth separation between the slower, inertia-dominated outer speed loop and the high-bandwidth inner PI current regulators. The integration of Space Vector PWM further verifies the capability to maximise DC bus utilisation and suppress harmonic distortion, making this a complete and robust control strategy.

5.2 Future Scope

Future enhancements to this control architecture could include:

- **Sensorless Control Integration:** Replacing the mechanical speed sensor with a Model Reference Adaptive System (MRAS) or Extended Kalman Filter (EKF) to estimate rotor speed purely from terminal voltages and currents.
- **Online Parameter Identification:** Implementing an adaptive algorithm to estimate the rotor resistance (R_r) in real-time, preventing IFOC detuning due to thermal variations during extended operation.
- **Field Weakening Operation:** Extending the control algorithm to dynamically attenuate i_{ds} when operating above base speed, allowing the motor to enter the constant-power region.

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